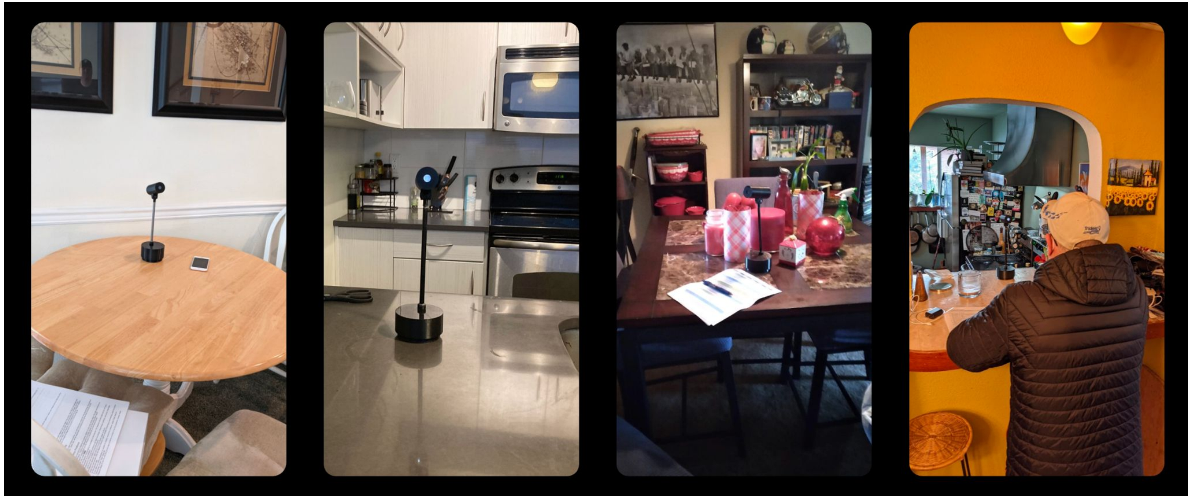
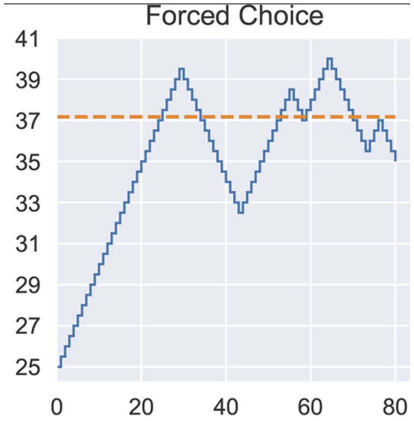
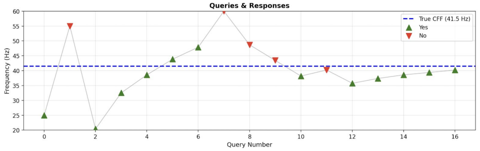
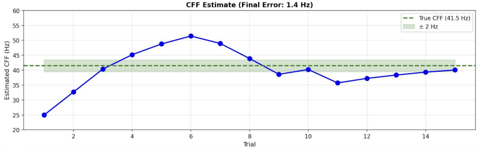
Adaptive Bernoulli Level Set Estimation for Efficient Critical Flicker Frequency Measurement with BEACON





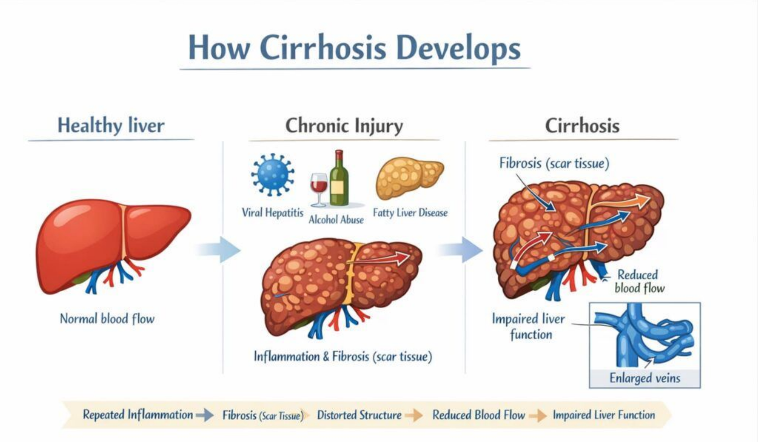




Andy Ye, Mentor: Richard Li

Chronic Liver Disease

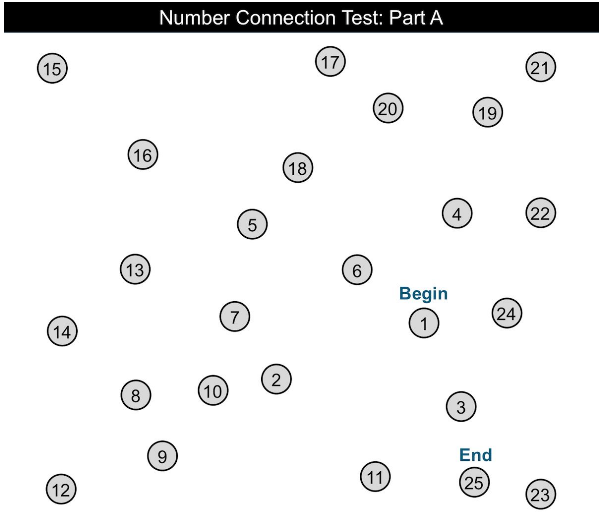
- General liver disease leads to **scarring of the liver**; **cirrhosis**
- Liver doctors tend to test cirrhosis, but cirrhosis causes **side effects**
- **One of these side effects deteriorates brain function**
  - Hepatic Encephalopathy



from: https://www.startstemcells.com/liver-cirrhosis-treatment.html

How to Measure for Hepatic Encephalopathy?

- Current tests are non-rigorous

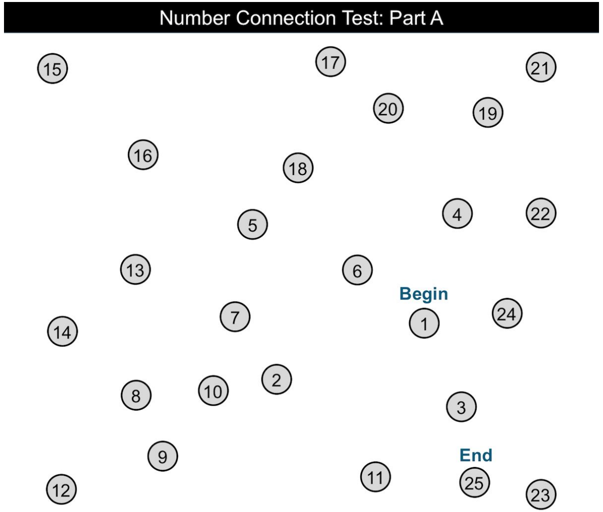


Number Connection Test

Flapping Hand Test

How to Measure for Hepatic Encephalopathy?

- Current tests are non-rigorous
- Emerging research in **critical flicker frequency** correlation

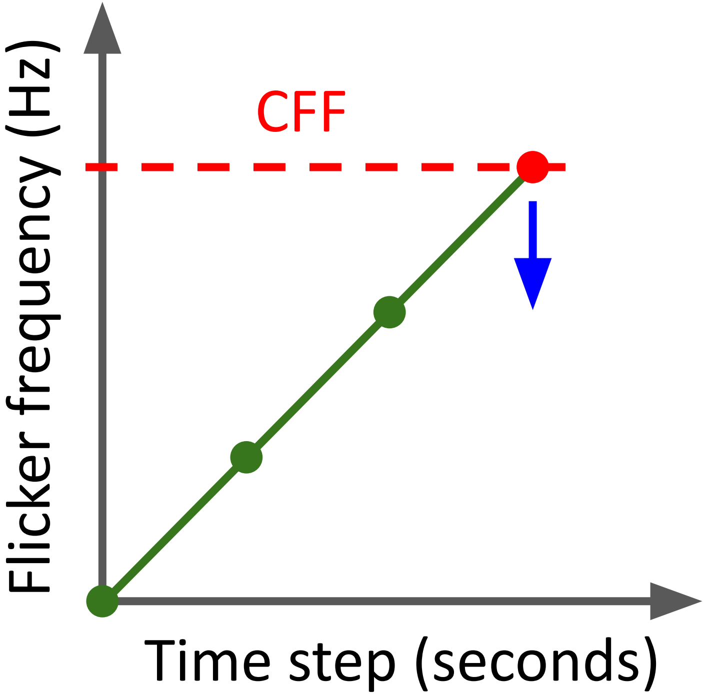


Number Connection Test

Flapping Hand Test

What is Critical Flicker Frequency (CFF)?

*from Richard Li**



Current Measurement Techniques

Lafayette Flicker Fusion System

Beacon

He’s in the room!

Beacon

Methods of Measuring CFF

- Method of Limits
- Staircase Method

From Lee Et. Al

Two Alternative Forced Choice (2AFC)

- This is standard in psychophysic measurements
- The user is presented with two lights; one flickering, and one not. Their job is to determine the flickering one
- This helps prevent biases
- This takes longer than just saying yes or no!

Methods of Measuring CFF

- Method of Limits
- Staircase Method

These take a long time! ~15 mins

From Lee Et. Al

Research Question

How can we ask the right questions to find out a person’s critical flicker frequency (CFF) in as little time and mental cost as possible?

What is Level Set Estimation (LSE)?

- **L = {x : f(x) = c}** … We just want to find the set of x's where f(x)=c.
- *"How fast does a bird have to run such that a coyote can't react to it?"*

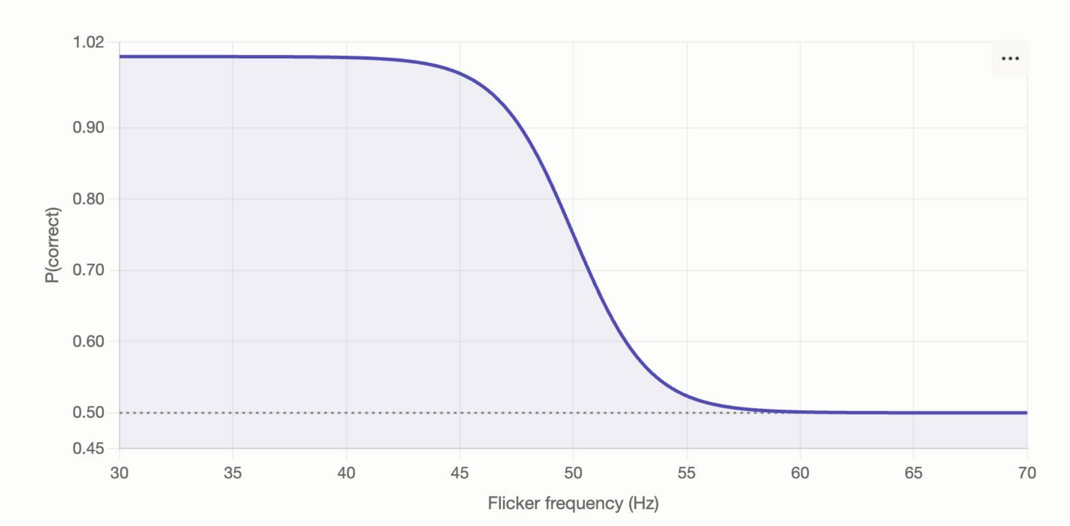
What is Level Set Estimation (LSE)?

- **L = {x : f(x) = c}** … We just want to find the set of x's where f(x)=c.
- *"How fast does a bird have to run such that a coyote can't react to it?"*
- ***"What flicker frequency (x) will user be at their threshold (c)?"***

What are we looking for? What's our c?

- The probability of them seeing the light flicker at their CFF is:

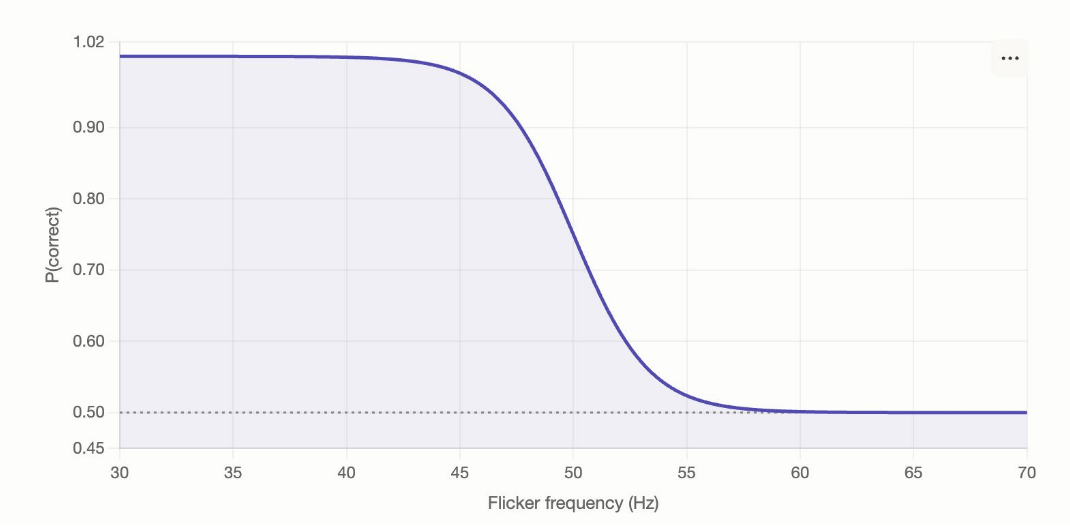
% Correct at CFF



What are we looking for? What's our c?

- The probability of them seeing the light flicker at their CFF is:

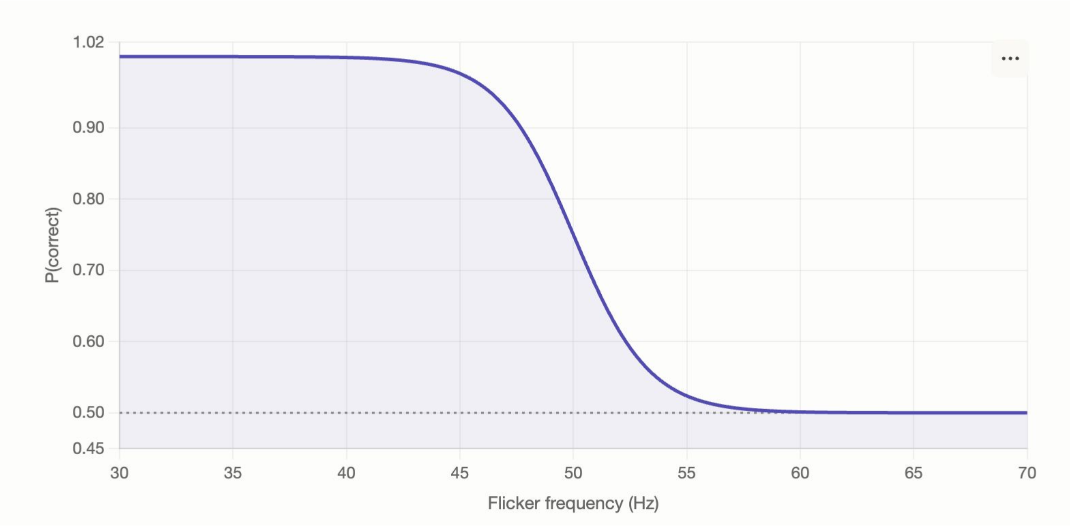
% Correct at CFF = Can see it: 50%



What are we looking for? What's our c?

- The probability of them seeing the light flicker at their CFF is:

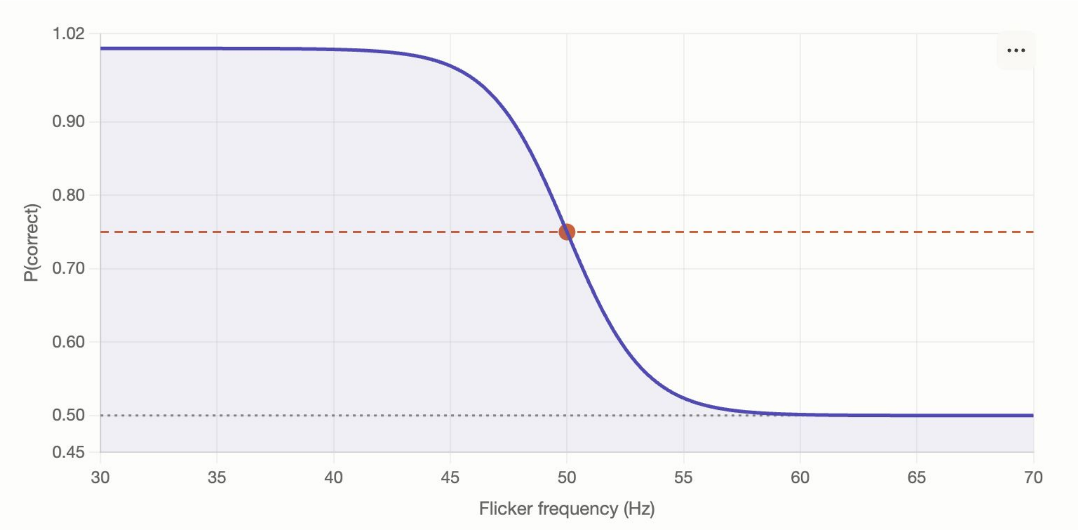
% Correct at CFF = Can see it: 50% + **Guessing: 25%**



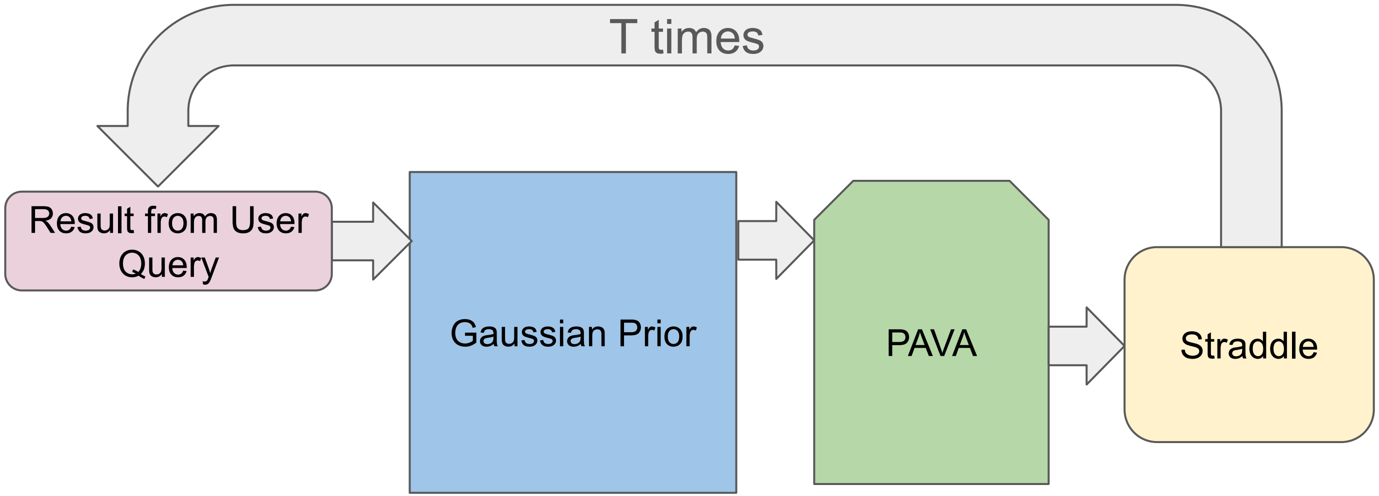
What are we looking for? What’s our c?

- The probability of them seeing the light flicker at their CFF is:

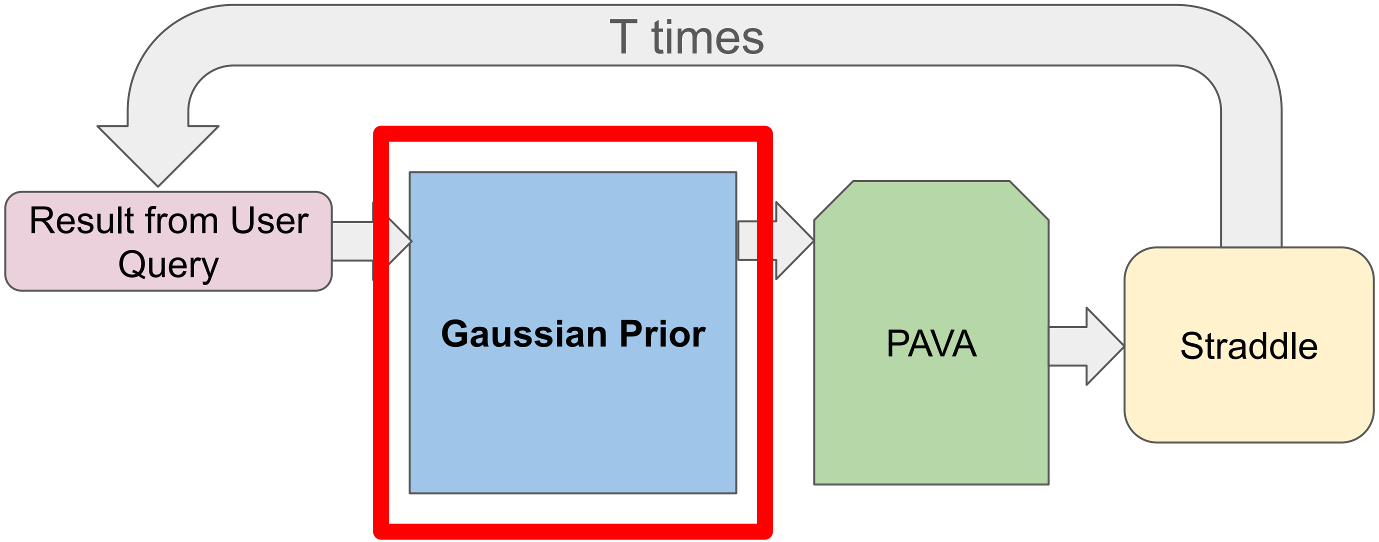
% Correct at CFF = Can see it: 50% + Guessing: 25% = **75%**



GP + MONO (GPM)


T times
Result from User Query
Gaussian Prior
PAVA
Straddle

GP + MONO (GPM)


T times
Result from User Query
Gaussian Prior
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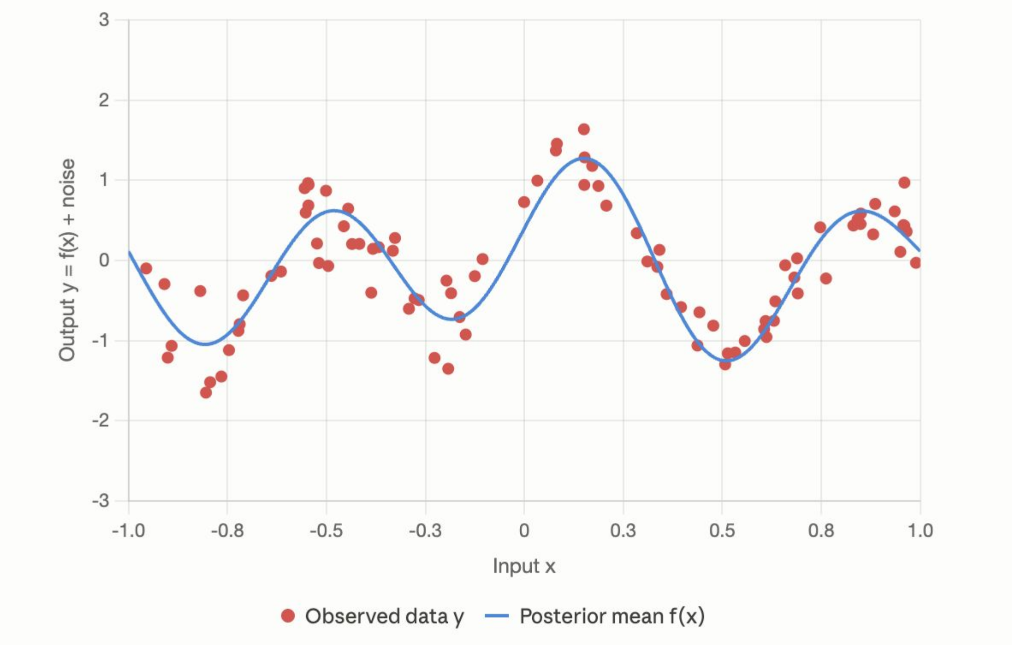
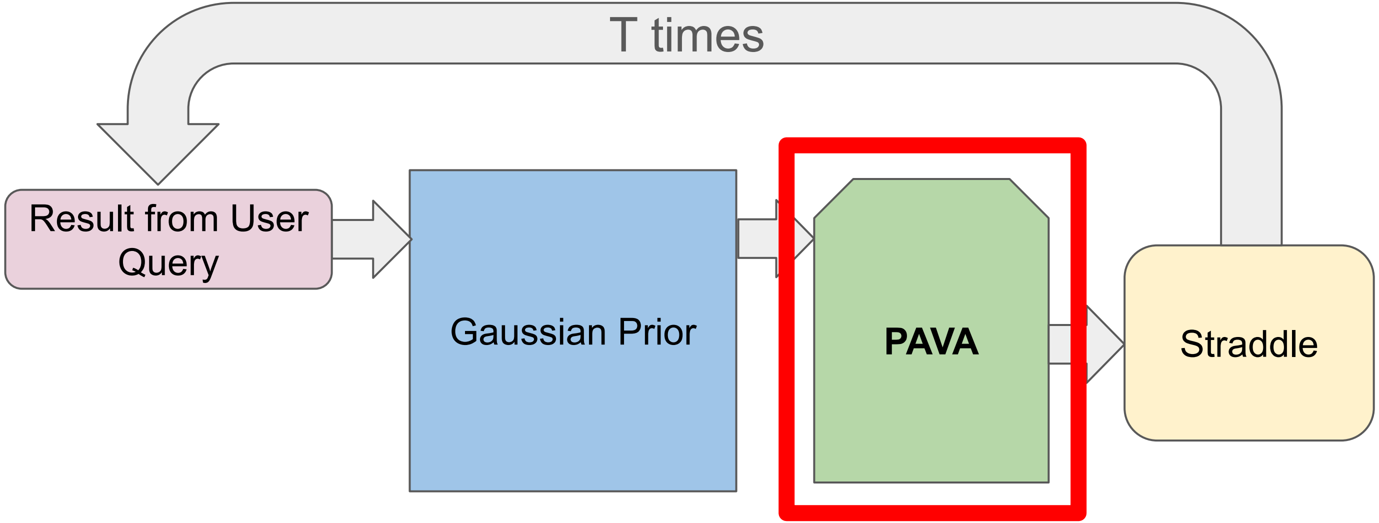
Gaussian Prior

- A Gaussian Prior (GP) is a probability distribution over **functions.**

A Function Fitting Observed Data



GP + MONO (GPM)



T times
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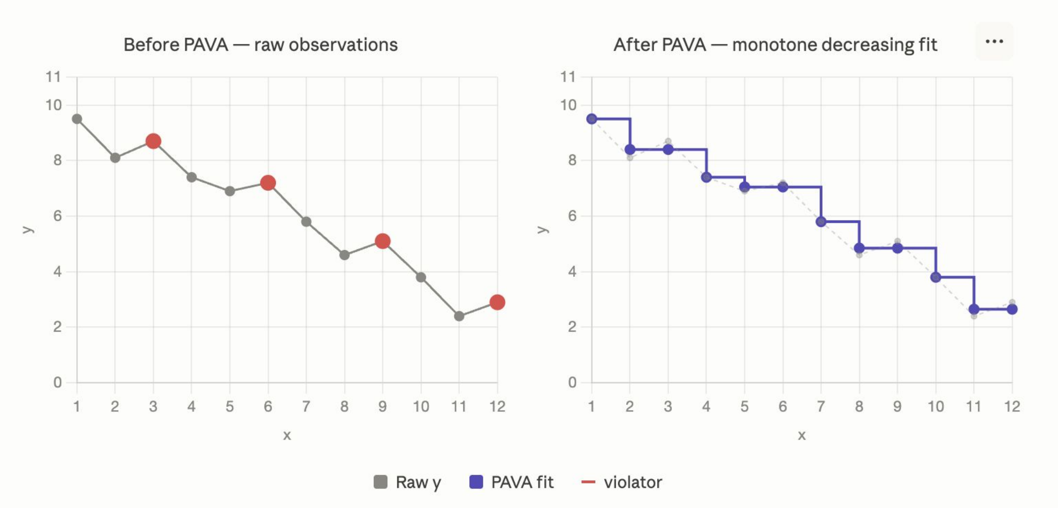
Pool Adjacent Violators Algorithm (PAVA)

- P(x) is **monotonically decreasing!**
- Enforces a **structural constraint**; course correction!
    - Due to 2AFC, our data can have a lot of guesswork
    - P({x,...}) as 0.95, **0.7,0.8,** 0.6 -> **PAVA to:** 0.95, **0.75,0.75**,0.6



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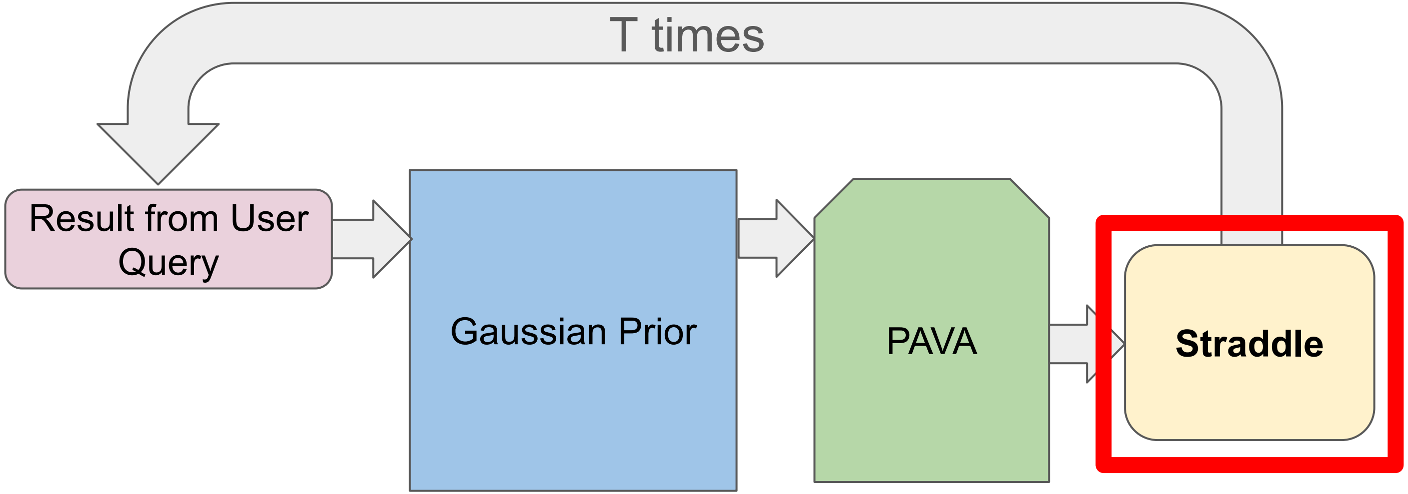
T times
Result from User Query
Gaussian Prior
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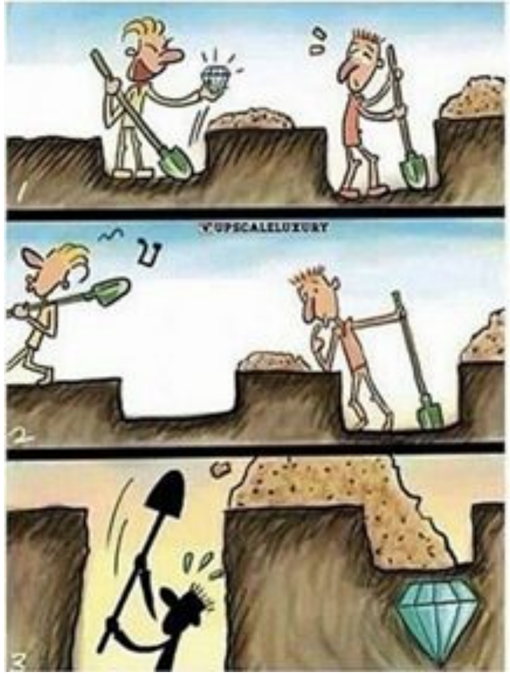
Straddle

Acquisition Functions

- An acquisition function chooses what **questions** to ask the user
  - *What frequency should I ask next?*
  - *What datapoint matters?*
- Exploration versus Exploitation

Quiz Machine: https://en.wikipedia.org/wiki/Quiz_machine



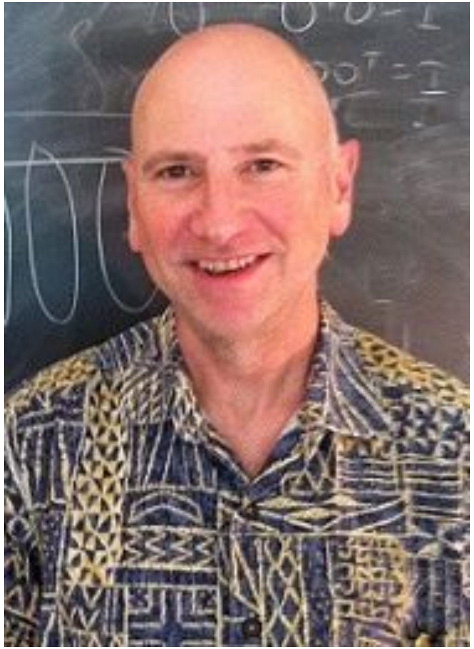
Straddle

- Level Set Estimator
- Built to use as few samples as possible
- Made to fit Gaussian Processes!
- “Straddles” the threshold

Active Learning For Identifying Function Threshold Boundaries

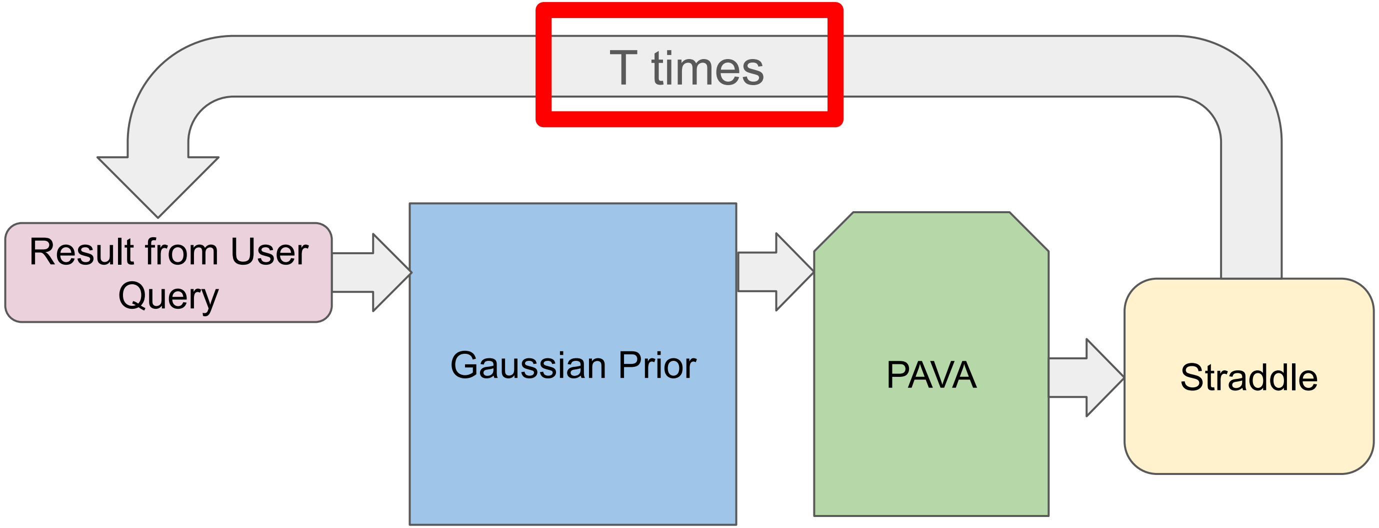
Brent Bryan, Robert C. Nichol, Christopher R Genovese, Jeff Schneider, Christopher J. Miller, Larry Wasserman

Advances in Neural Information Processing Systems 18 (NIPS 2005)



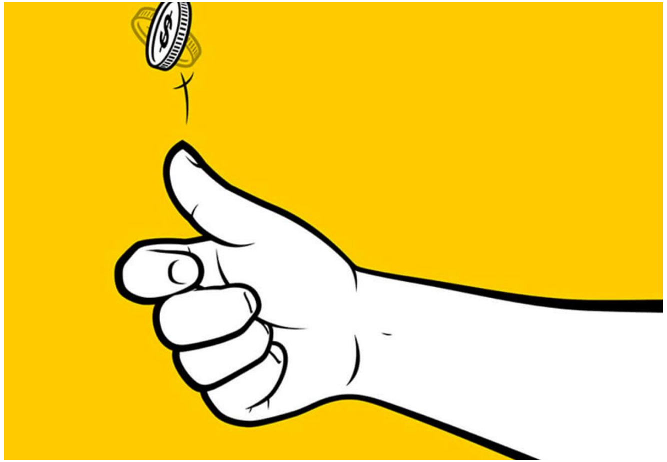
Larry Wasserman, CMU

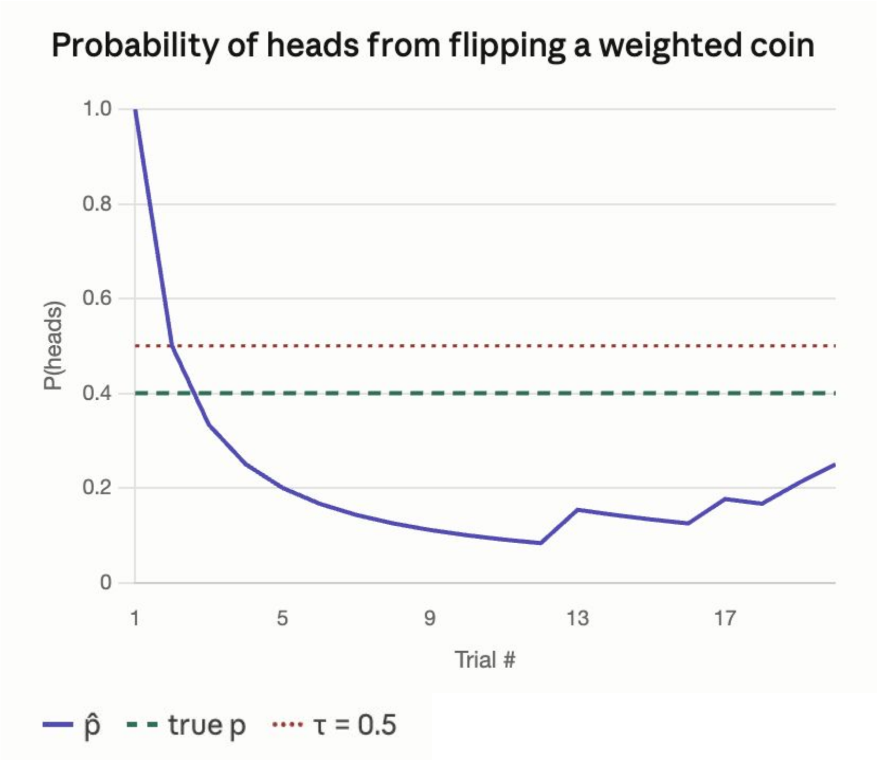
GP + MONO (GPM)


T times
Result from User Query
Gaussian Prior
PAVA
Straddle

Bernoulli LSE?

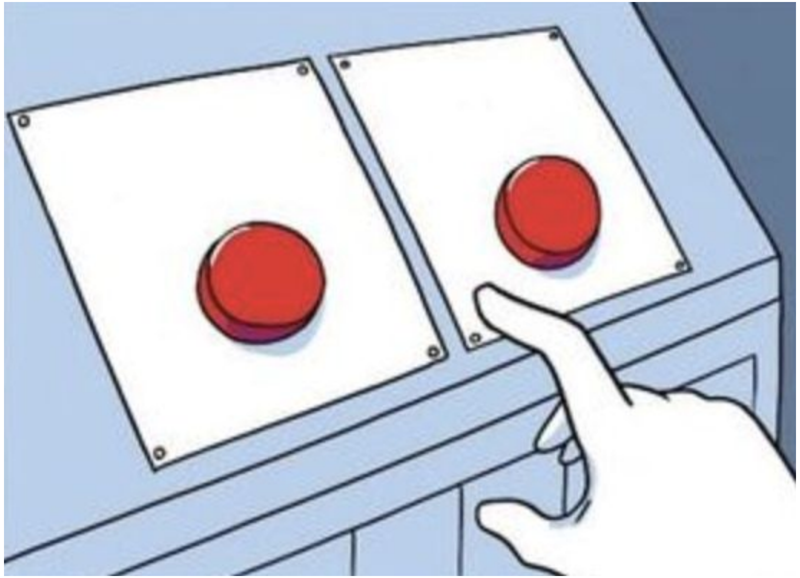
- What is a Bernoulli variable?
- Data becomes very jumpy!
- Harder to find correlations





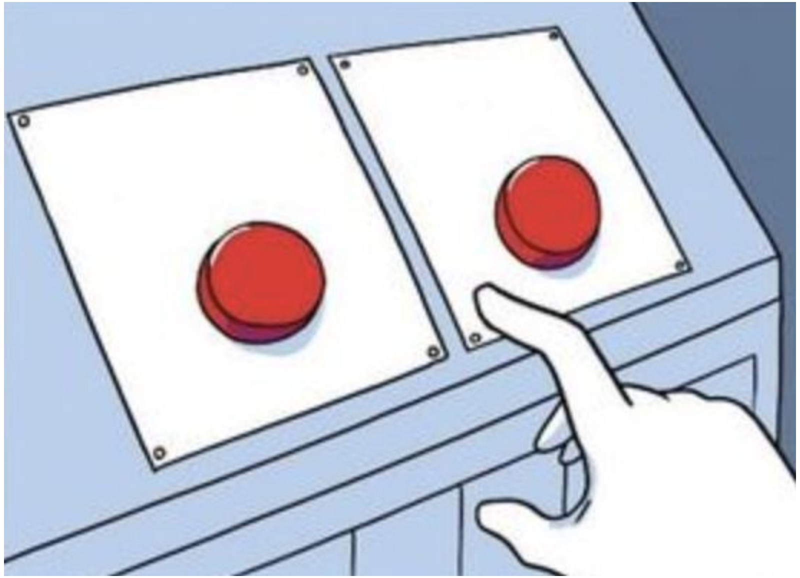
Noisy Data Can Throw Us Off!

- Because the user is guessing when they can't tell, this makes for **noisy data!**

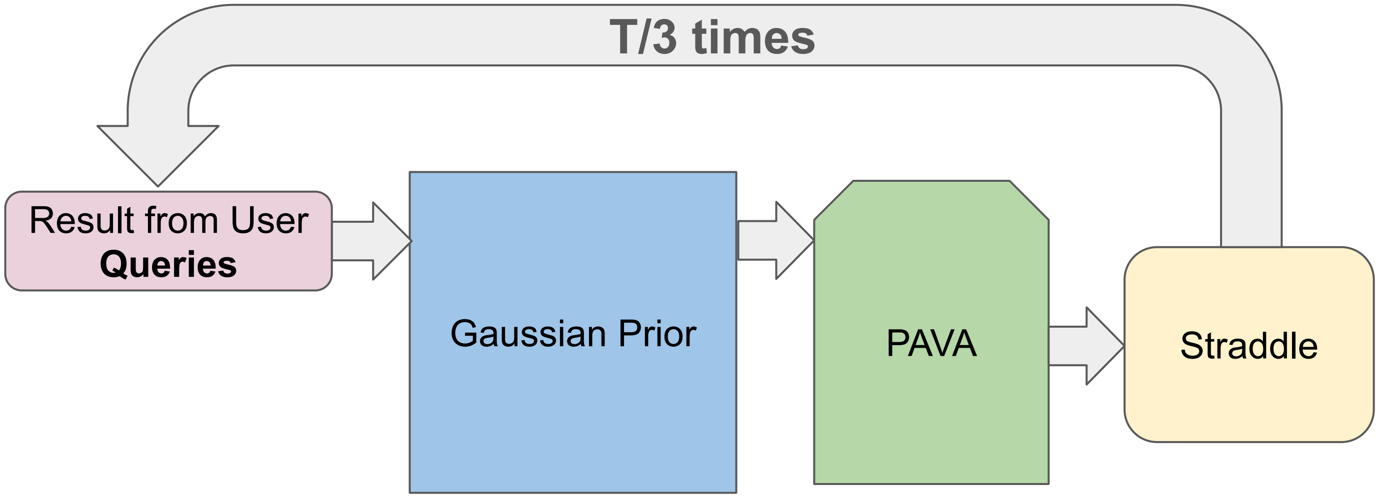


Noisy Data Can Throw Us Off!

- Because the user is guessing when they can't tell, this makes for **noisy data!**
- **Solution**: query each frequency 3 times



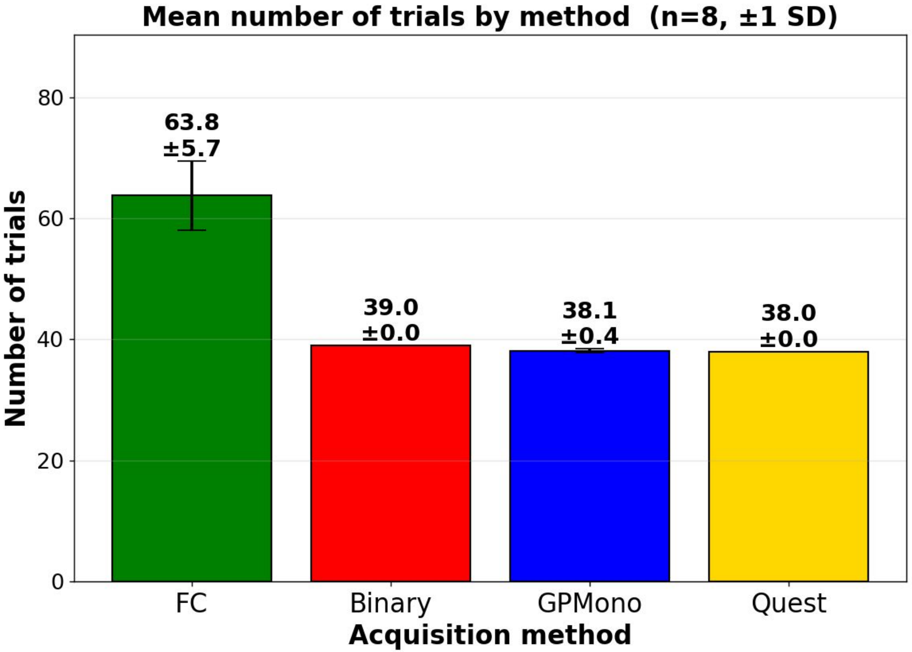
GP + MONO (GPM)


T/3 times
Result from User **Queries**
Gaussian Prior
PAVA
Straddle

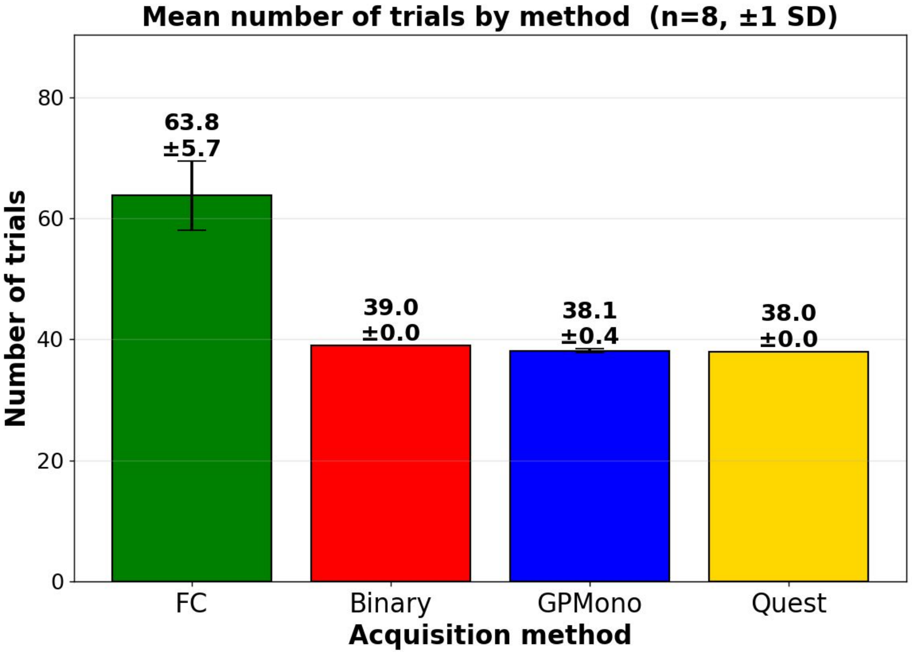
Testing Protocol

1. Collected data from 8 participants
    a. Binary Search (LSE)
    b. Quest+ (LSE)
    c. GP+Mono (LSE)
    d. Staircase 2AFC (Baseline)
2. Generated Latin Square for Ordering
3. Solicited qualitative feedback

Results


Mean number of trials by method (n=8, ±1 SD)
Number of trials
80
60
40
20
0
63.8
±5.7
39.0
±0.0
38.1
±0.4
38.0
±0.0
FC
Binary
GPMono
Quest
Acquisition method

Results


Mean number of trials by method (n=8, ±1 SD)
Number of trials
0
20
40
60
80
63.8
±5.7
39.0
±0.0
38.1
±0.4
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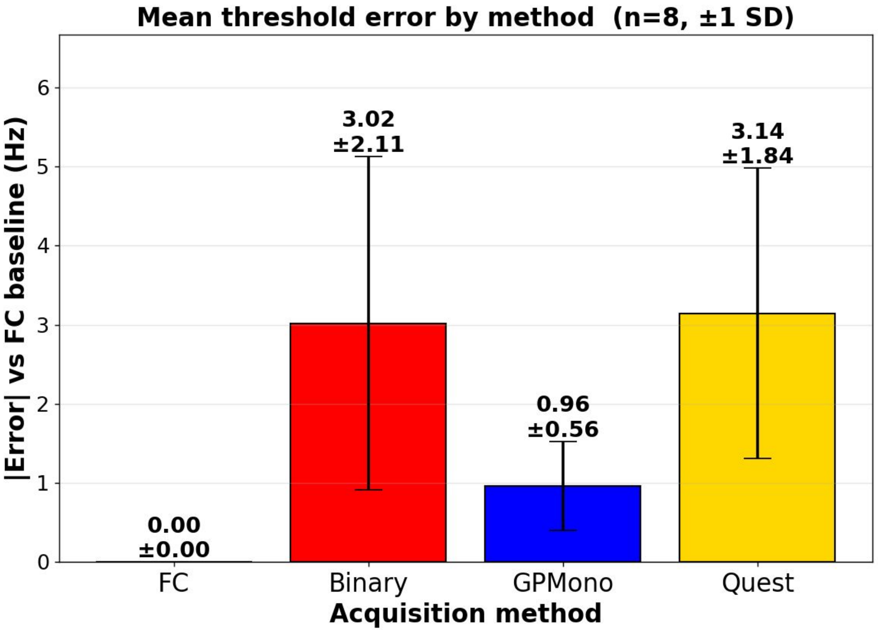
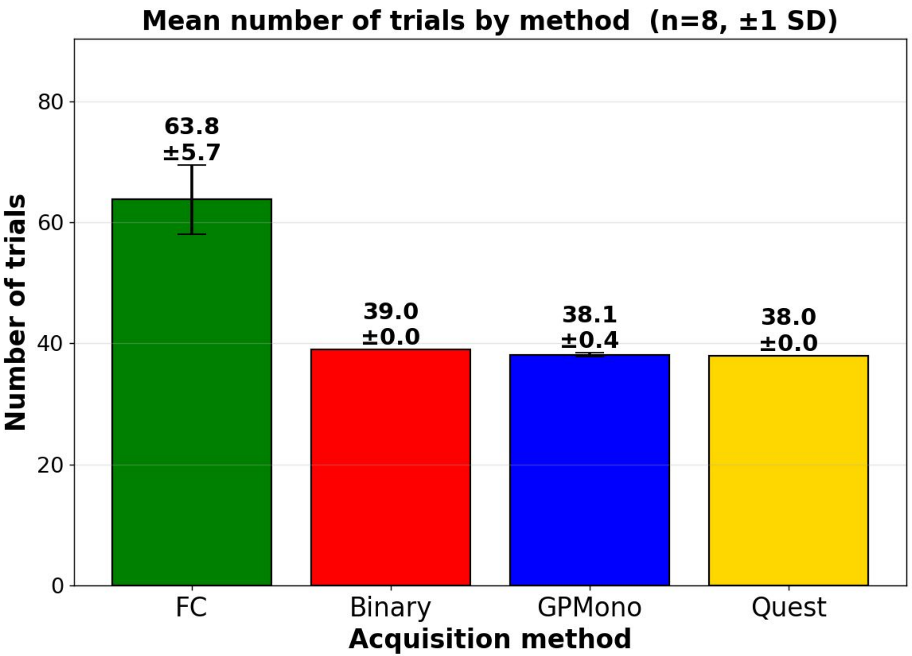

Mean threshold error by method (n=8, ±1 SD)
|Error| vs FC baseline (Hz)
0
1
2
3
4
5
6
0.00
±0.00
3.02
±2.11
0.96
±0.56
3.14
±1.84
FC
Binary
GPMono
Quest
Acquisition method

Results


Mean number of trials by method (n=8, ±1 SD)
Number of trials
0
20
40
60
80
63.8
±5.7
39.0
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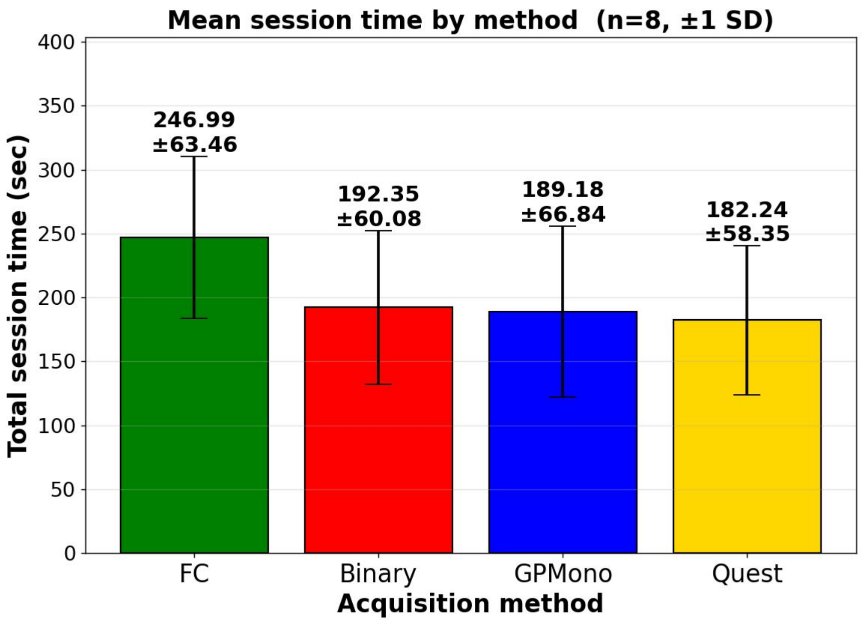

Mean session time by method (n=8, ±1 SD)
Total session time (sec)
0
50
100
150
200
250
300
350
400
246.99
±63.46
192.35
±60.08
189.18
±66.84
182.24
±58.35
FC
Binary
GPMono
Quest
Acquisition method

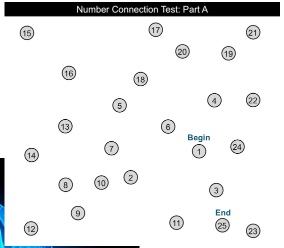
General Observations and User Feedback

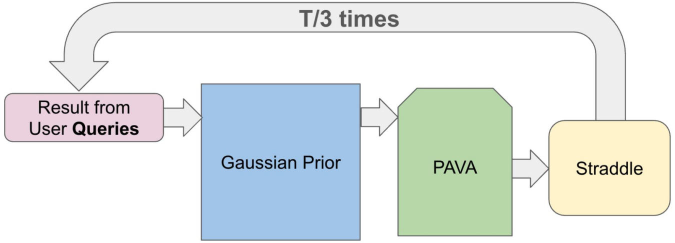
- Users critical flicker frequencies have **variance**
- Peripheral flicker frequency
- Users tried to ‘game’ the system
    - *“I think the dimmer one is the correct answer”*
    - *“I generally chose the brighter one”*
    - *“I looked for small patterns when switching between the two”*

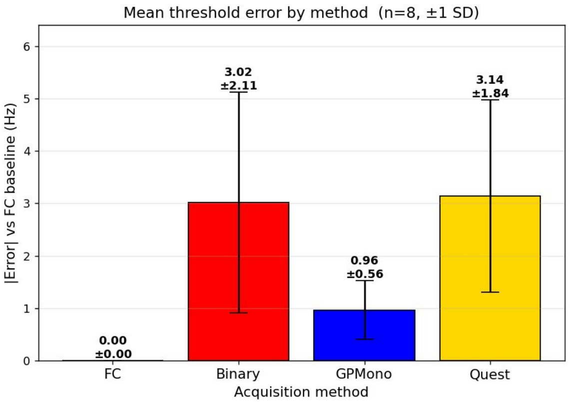
Limitations and Future Work

- Testing-retest reliability across methods
- More hyperparameter tuning
- Need for more tests on people with hepatic encephalopathy
    - Current sample is biased, all college students!
- More information on CFF variance per user

Summary







Thank you!